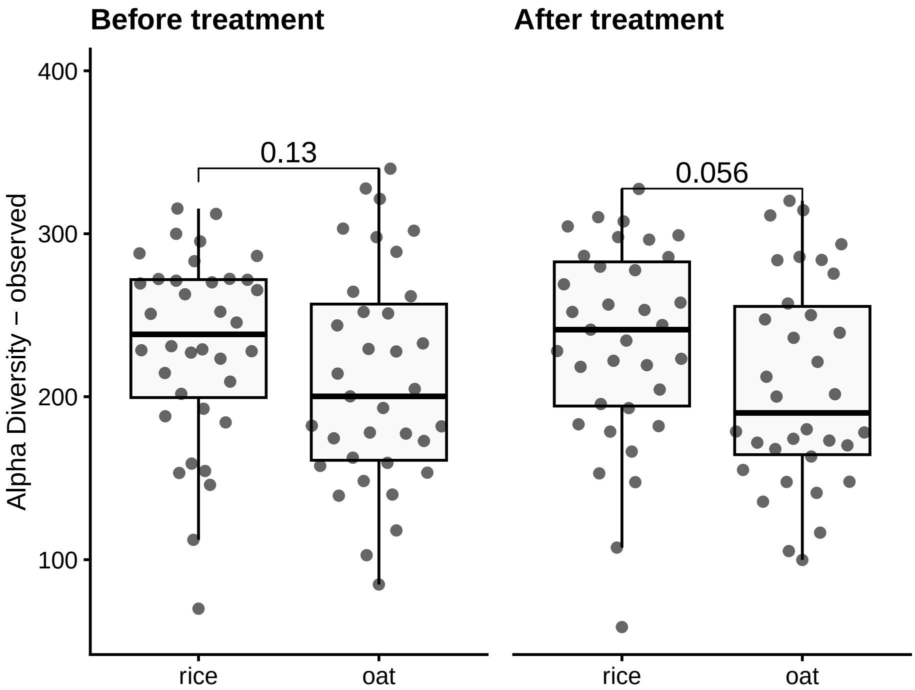
Alpha Diversity

Muluh

1 Group-wise comparisons

- **Index:** observed
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



1.1 Table of the top findings

1.1.1 Across-group

1.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

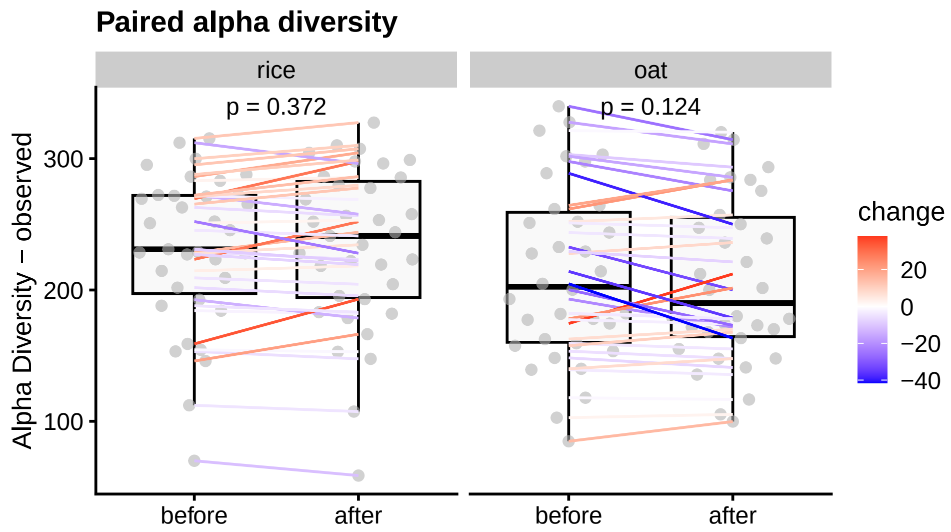
group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	211.1	231.6	0.13	0.13

1.1.1.2 After treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
oat	rice	207.1	233.2	0.17	0.056



1.1.2 Within-group

1.1.2.1 Paired treatment

- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
rice	231.6	233.2	0.13	0.372
oat	211.1	207.1	0.04	0.124

1.2 LMM

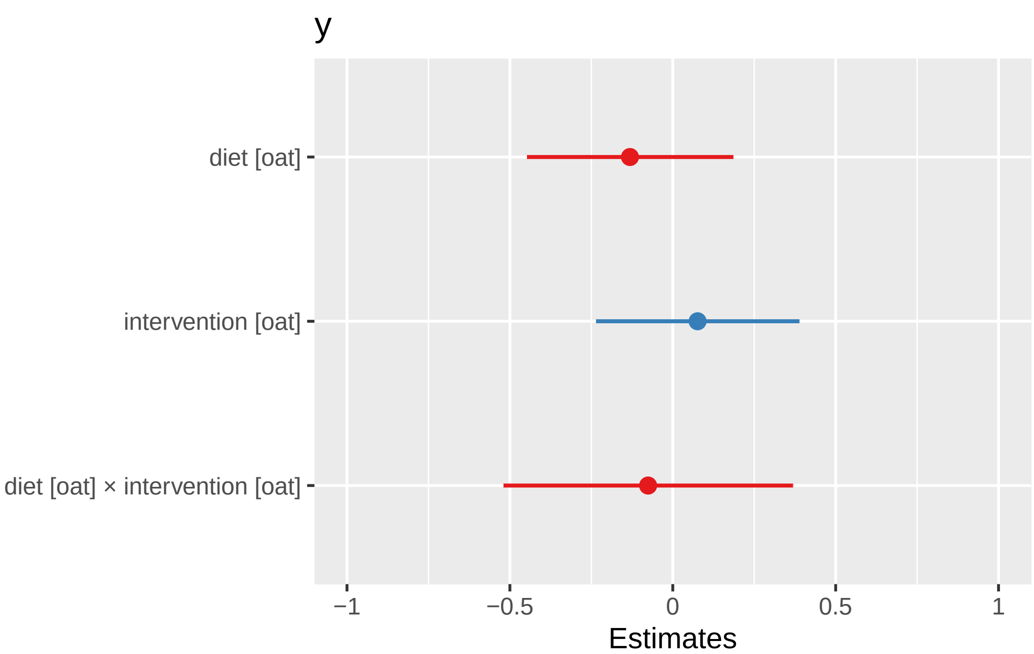
- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**
- Formula used: ~ **diet * intervention + (1 | id)**

Characteristic	before			after		
	Beta	95% CI[1]	p-value	Beta	95% CI[1]	p-value
(Intercept)	151	67, 235	<0.001	143	59, 228	0.001
meal_group			0.4			0.3
1	—	—		—	—	
2	32	-11, 75	0.14	28	-16, 72	0.2
3	28	-15, 71	0.2	30	-14, 73	0.2
4	10	-32, 53	0.6	-3.6	-47, 39	0.9
age	1.0	-0.44, 2.5	0.2	1.2	-0.24, 2.7	0.10

[1]CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI[1]	p-value
(Intercept)	147	84, 209	<0.001
meal_group			0.6
1	—	—	
2	28	-16, 71	0.2
3	28	-15, 71	0.2
4	2.4	-27, 32	0.9
age	1.2	0.14, 2.2	0.025

[1]CI = Confidence Interval



2 Alternate analyses

- **Beta (Estimate)**: Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI**: Range where the true beta is likely to fall, reflecting precision.
- **p-Value**: Tests if beta differs from zero; p<0.05p < 0.05p<0.05 suggests significance.

3 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(17): sample id ... group observed_log2
reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```